

Lorentzian propinquity convergence on truncated $SU(2) \times U(1)_T$ Krein spectral triples

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Abstract

We construct a Lorentzian analog of the Latrémolière quantum Gromov–Hausdorff propinquity, restricted to the Krein-positive state space, and prove the corresponding convergence theorem on truncated Krein spectral triples over the compact-temporal Wick-rotated manifold $S^3 \times S_T^1$. Concretely, let $\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ be the chirality-doubled Camporesi–Higuchi spinor space tensored with the N_t -mode Fourier truncation of $L^2(S_T^1)$, with fundamental symmetry $J_L = J_L^{\text{spat}} \otimes I_{N_t}$ at the Bizi–Brouder–Besnard [3] classification $(m, n) = (4, 6)$ in the Peskin–Schroeder chiral basis, and let $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t})$ be the Lorentzian Dirac per van den Dungen 2016 [32] Proposition 4.1, with ∂_t Fourier-diagonal anti-Hermitian. Let $\mathcal{O}_{n_{\max}, N_t, T}^L$ be the truncated Lorentzian operator system built from chirality-doubled scalar 3- Y spatial multipliers tensored with the momentum-diagonal temporal multiplier subalgebra. On the Krein-positive subspace $\mathcal{K}^+ := \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$ we obtain a standard metric spectral triple $\mathcal{T}_{n_{\max}, N_t, T}^+$ to which Latrémolière 2017/2023 [16] machinery applies verbatim, and define the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity $\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+)$. The main theorem reads

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0,$$

with $C_3^{\text{joint}} \leq 1$ asymptotically tight (inherited from Paper 38 [36] Lemma L3 via a structural identity for the joint commutator), and $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ the joint mass-concentration moment ($SU(2)$ factor inherits the $4/\pi$ asymptote of Paper 38 Lemma L2; $U(1)$ factor inherits the standard Fejér-on-the-circle rate). The proof is a five-lemma argument L1', L2, L3, L4, L5 transferred from Paper 38 with one substantive structural simplification: in the momentum-polynomial convention for temporal multipliers, the joint commutator $[D_L, a_s \otimes a_t]$ reduces *bit-exactly* to $i[D_{\text{GV}}, a_s] \otimes a_t$, so the temporal direction contributes nothing to the joint Lipschitz comparison; the joint C_3^{joint} equals the spatial $C_3^{SU(2)}$ verbatim. The Riemannian limit at $N_t = 1$ reduces Λ^L to Paper 38's bound bit-exactly (load-bearing falsifier). To our knowledge this is the first Lorentzian propinquity convergence theorem on truncated Krein spectral triples in the published math.OA literature. Honest scope: the construction is a *weak-form* propinquity on the $\mathcal{K}^+$ Hilbert-space restriction; a *strong-form* Latrémolière-style metric on Krein-signature spectral triples without $\mathcal{K}^+$ restriction remains an open NCG-math problem. The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] lives on a categorically different mathematical object (pre-length spaces with causal diamonds, not operator-algebraic spectral triples).

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1 Introduction

This paper closes the convergence-theorem leg of the Lorentzian-extension program for truncated GeoVac spectral triples, following the math.OA quartet Papers 38–40 [36–38] (Riemannian propinquity convergence on $SU(2)$, on tensor products of $SU(2)$ triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) and the Lorentzian extensions Papers 42–44 [39–41] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff).

The dominant blocker for transferring Latrémolière propinquity [16, 17] to the Lorentzian setting is the absence of a published metric on Krein-signature spectral triples in their own right: Latrémolière’s construction crucially uses the operator-norm Lipschitz seminorm via the Dirac commutator $\| [D, \cdot] \|_{\text{op}}$, well-defined on Hilbert space but requiring an indefinite-inner-product analog on a Krein space (with the appropriate Krein-self-adjointness of D_L per van den Dungen 2016 [32] / BBB [3]). We sidestep this gap by working on the Krein-positive subspace $\mathcal{K}^+$, where the Krein product reduces to a positive-definite Hilbert inner product, and the standard Latrémolière machinery applies verbatim. We call this the *$\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity*.

1.1 What this paper does

We work on the compact-temporal Wick-rotated geometry $S^3 \times S_T^1$, with Riemannian metric $ds^2 = d\chi^2 + \sin^2 \chi d\Omega^2 + d\tau^2$ at canonical circumference $T = 2\pi$. The Krein structure J_L is preserved as an *algebraic* object (operator system plus symmetry, in the BBB (4, 6) classification) but no Lorentzian-causal-structure claim is made; the closed timelike-curve obstruction (Geroch [11]) is acknowledged explicitly via the Wick-rotated convention. The construction proceeds as a five-lemma argument transferred from Paper 38:

- (L1′) The Lorentzian operator-system substrate $\mathcal{O}_{n_{\max}, N_t, T}^L$ of Paper 44 [41], with propagation number 2 on the achievable envelope and trivial Krein-positivity at the operator-multiplier level.
- (L2) A joint central spectral Fejér kernel $K^{\text{joint}} = K_{n_{\max}}^{\text{SU}(2)} \otimes K_{N_t, T}^{U(1)}$ on $SU(2) \times U(1)$ with factorized Plancherel symbol and joint cb-norm $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max} + 1)$, via the Bożejko–Fendler central-multiplier equality [4, 26] on the amenable compact group product.
- (L3) The joint Lichnerowicz / Lipschitz comparison $\|[D_L, a]\|_{\text{op}} \leq C_3^{\text{joint}} \|\nabla^{\text{joint}} a\|_{\infty}$ with $C_3^{\text{joint}} \leq 1$, via the structural identity $[D_L, a_s \otimes a_t] = i[D_{\text{GV}}, a_s] \otimes a_t$ which makes the temporal direction contribute nothing to the joint commutator.
- (L4) A joint Berezin reconstruction $B_{n_{\max}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\max}, N_t, T}^L$ with positivity, contractivity, approximate identity (rate $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$), and L3 compatibility.
- (L5) Assembly of the propinquity tunneling pair $(B^{\text{joint}}, P^{\text{joint}})$ in the Latrémolière weak-form framework on $\mathcal{K}^+$, giving the main bound $\Lambda^L \leq C_3^{\text{joint}} \cdot \gamma^{\text{joint}} \rightarrow 0$.

1.2 Honest scope and named gaps

The construction is at the $\mathcal{K}^+$ -**restricted weak-form** level. The four named gaps to the strong-form construction are:

- (G1) *Strong-form Lorentzian propinquity*. A Latrémolière-style metric on Krein-signature spectral triples without $\mathcal{K}^+$ restriction is not in the published literature as of May 2026. The required ingredient is an indefinite-inner-product analog of the operator-norm Lipschitz seminorm with appropriate behavior under Krein-self-adjoint D_L . This is a multi-month NCG-math problem not addressed here.
- (G2) *De-compactification limit $T \rightarrow \infty$* . Our construction is at compact temporal radius T canonical $T = 2\pi$ (the BW modular period). The non-compact $\mathbb{R}_t$ limit re-introduces the joint compact $\times$ non-compact issue. **The norm-resolvent / spectral level of this limit is closed in Paper 47 [43] § 4–5 via a two-rate hybrid composite: the inner arrow is parametric propinquity stability of the present construction at coupled cells $(n_{\max}, N_t, T(N_t))$ along an admissible scaling $T(N_t) \rightarrow \infty$ with $T(N_t)/N_t \rightarrow 0$, and the outer arrow is norm-resolvent convergence of the Lorentzian Dirac on $S^3 \times S^1_T$ to the Lorentzian Dirac on $S^3 \times \mathbb{R}_t$ with explicit exponential tail $O(e^{-|\operatorname{Im} z| T/2})$ on compact-support subspaces.** The *metric / propinquity* level of this limit on the non-compact carrier remains open (it requires a non-compact extension of Latrémolière’s propinquity, which is not in the published literature as of May 2026): named G2-metric or equivalently G1-genuine on the non-compact substrate (Sprint L3c- β / L3e in our internal taxonomy).
- (G3) *Cross-manifold extensions* of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ are blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (four-layer Coulomb/HO asymmetry from Paper 24 [34] § V) and are not addressed.
- (G4) *Inner-factor calibration data* (the Higgs / Yukawa selection question of the Connes–Marcolli SM construction) is orthogonal to the present convergence theorem.

1.3 Related work

Latrémolière’s quantum Gromov–Hausdorff propinquity [17] (Trans. AMS 2018) and the metric-spectral-triple weak-form extension [16] (Adv. Math. 2023) supply the convergence framework we work within. The most recent Riemannian-side continuity result is Latrémolière 2025 [9], treating analytic paths of Riemannian metrics; none of these constructions extend to Krein signature.

Connes–van Suijlekom 2021 [6] establishes the Gromov–Hausdorff convergence of spectral truncations on the circle S^1 and explicitly defers convergence on other manifolds (“elsewhere” on three occasions). The subsequent flat-structure program [8, 13, 14, 18, 30] covers $\mathbb{T}^d$, polynomial-growth groups, and the Berezin–Toeplitz S^2 , with universal $4/\pi$ asymptote on compact Lie groups established in Paper 40 [38] (also adapted to $SU(2)$ in Paper 38 [36]). All of these are strictly Riemannian.

The Lorentzian Krein-spectral-triple framework derives from Strohmaier [28] (axiomatic foundation), Franco–Eckstein [10] (algebraic causality), van den Dungen 2016 [32] (Krein spectral triples and fermionic action; the Wick-rotation lift used here), Bizi–Brouder–Besnard 2018 [3] ((m, n) classification of indefinite signatures used at (4, 6) West-coast in the present paper), Devastato–Lizzi–Martinetti [7], and the recent $SU(1, 1)$ construction [33]; none of these works define a metric on the algebra of operator systems extracted from a Krein triple. The closest algebraic alternative is the Nieuviarts twist morphism [22, 23], which produces pseudo-Riemannian spectral triples from Riemannian ones via a twist on twisted spectral triples; this is not a metric

and moreover (as noted in Paper 42 § 3.2) is restricted to even-dimensional manifolds, so does not apply to S^3 at KO-dimension 3.

The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via ϵ -nets of causal diamonds; a downstream pure-metric extension via the intrinsic timed-Hausdorff distance appears in Che–Perales–Sormani [21]. This is conceptually adjacent (all are “Lorentzian Gromov–Hausdorff-type convergence”) but lives on a categorically different mathematical object: pre-length spaces with causal-diamond / timed-Hausdorff geometry, not operator-algebraic spectral triples. The two families of constructions are not in competition; neither implies the other.

1.4 Outline

§ 2 fixes conventions: Krein space, Lorentzian Dirac, operator system, joint Fejér kernel, Krein-positive subspace, and the definition of the $\mathcal{K}^+$ -restricted weak-form propinquity. § 3 reviews the $L1'$ operator-system substrate (Paper 44). § 4 proves the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3). § 5 states and proves the main theorem (L5 propinquity assembly). § 6 records the numerical verification panel and the bit-exact Riemannian-limit recovery at $N_t = 1$. § 7 discusses the four-witness Wick-rotation connection (Paper 42/43), records the open questions (strong-form, de-compactification, cross-manifold), and positions the work against Mondino–Sämman. Appendix A records the symbolic Plancherel calculation for the joint Fejér symbol. Appendix B records the numerical panel data.

2 Preliminaries

2.1 Compact-temporal manifold and Wick-rotated metric

The compact-temporal Lorentzian / Wick-rotated manifold is $S^3 \times S_T^1$, with $S^3 = \mathrm{SU}(2)$ the round unit three-sphere (spatial slot) and $S_T^1 = \mathbb{R}/T\mathbb{Z}$ the temporal circle of circumference T . The Riemannian / Wick-rotated metric is

$$ds^2 = d\chi^2 + \sin^2 \chi d\Omega_2^2 + d\tau^2, \quad (1)$$

with $\chi \in [0, \pi]$ the polar angle on S^3 , $d\Omega_2^2$ the round metric on the $\mathbb{S}^2$ base of the Hopf fibration, and $\tau \in \mathbb{R}/T\mathbb{Z}$ the temporal coordinate. We treat the Krein structure J_L defined below as an *algebraic* object that encodes the BBB (4,6) chirality grading without claiming a Lorentzian-causal interpretation of the underlying metric: Geroch’s no-CTC obstruction [11] prevents a Lorentzian metric on $S^3 \times S_T^1$ in which S_T^1 is timelike, so the present construction operates on the Wick-rotated geometry and lifts the Krein structure as indefinite-inner-product datum (per Strohmaier [28], vdD [32], BBB [3]).

The canonical choice $T = 2\pi$ matches the Bisognano–Wichmann [2] modular period at unit surface gravity; this is consistent with the four-witness arc closure of Paper 43 [40]. The proof of the main theorem holds for any $T > 0$; explicit dependence on T enters the $U(1)$ factor of the joint mass-concentration moment as $O(T/N_t)$.

2.2 Krein space and fundamental symmetry

Per Paper 43 [40]§ 2 / Paper 44 [41]§ 2, the Krein space is

$$\mathcal{K}_{n_{\max}, N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}, \quad (2)$$

where $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ is the chirality-doubled Camporesi–Higuchi [5] spinor Hilbert space truncated at the $n_{\max}$ -th Fock shell, of dimension

$$\dim \mathcal{H}_{\mathrm{GV}}^{n_{\max}} = \frac{2}{3} n_{\max}(n_{\max} + 1)(n_{\max} + 2), \quad (3)$$

and $\mathbb{C}^{N_t}$ is the N_t -mode Fourier truncation of $L^2(S_T^1)$. The fundamental symmetry is

$$J_L = J_L^{\text{spat}} \otimes I_{N_t}, \quad (4)$$

with $J_L^{\text{spat}} = \gamma^0$ the chirality-swap in the Peskin–Schroeder chiral basis, $(J_L^{\text{spat}})^2 = +I$.

The BBB (m, n) classification at signature $(s, t) = (3, 1)$ West-coast assigns $(m, n) = (s + t, t - s \bmod 8) = (4, 6)$ via BBB [3] Table 3, fixing $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$. These signs were verified bit-exact at finite cutoff in Paper 43 § 5 [40].

2.3 Lorentzian Dirac operator

The Lorentzian Dirac, per vdD 2016 [32] Prop. 4.1 applied at signature $(3, 1)$, reads

$$D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t}), \quad (5)$$

with ∂_t Fourier-diagonal anti-Hermitian on $\mathbb{C}^{N_t}$: in the momentum eigenbasis $\{|k\rangle : k \in \mathbb{Z}, |k| \leq K_{\text{max}}\}$ with $K_{\text{max}} := \lfloor (N_t - 1)/2 \rfloor$ for odd N_t or $N_t/2$ for even N_t ,

$$\partial_t = i \text{diag}(\omega_k), \quad \omega_k = 2\pi k/T, \quad (6)$$

so $\partial_t^\dagger = -\partial_t$. The spatial part D_{GV} is the truthful Camporesi–Higuchi spatial Dirac on $\mathcal{H}_{\text{GV}}^{n_{\text{max}}}$, diagonal in the spinor basis with eigenvalue $\chi(n+1/2)$ where $\chi \in \{+1, -1\}$ is the spinor chirality and n the Fock-shell index. Krein-self-adjointness $D_L^\times := J_L D_L^\dagger J_L = D_L$ is verified by direct computation: $\gamma^{0\dagger} = \gamma^0$ (Hermitian chirality-swap), $\{\gamma^0, D_{\text{GV}}\} = 0$ in the chiral basis, $\partial_t^\dagger = -\partial_t$, and $J_L^2 = +I$ combine to give the stated identity. The sign $i^t = +i$ in (5) is forced by this requirement (an opposite sign would give an anti-Krein-self-adjoint D_L); see Paper 43 § 3 [40] for the explicit derivation.

2.4 Truncated Lorentzian operator system

Per Paper 44 [41], the truncated Lorentzian operator system is the pure-tensor span

$$\mathcal{O}_{n_{\text{max}}, N_t, T}^L := \text{span}_{\mathbb{C}}\{M_{N, L, M}^{\text{spat}} \otimes M_p^{\text{temp}} : (N, L, M) \in \mathcal{I}_{n_{\text{max}}}^{\text{spat}}, p \in \{0, 1, \dots, N_t - 1\}\}, \quad (7)$$

where each spatial multiplier $M_{N, L, M}^{\text{spat}} = \text{blkdiag}(W_{N, L, M}, W_{N, L, M})$ is the chirality-doubled lift of the Avery–Wen–Avery Weyl-sector multiplier $W_{N, L, M}$ (the Clebsch–Gordan / Avery 3- Y integral construction of Paper 32 [35] § III), and the temporal multipliers M_p^{temp} are momentum-polynomial diagonal:

$$M_p^{\text{temp}} = \text{diag}(\omega_k^p)_{|k| \leq K_{\text{max}}}. \quad (8)$$

Vandermonde invertibility at N_t distinct frequencies ω_k ensures that the span (7) coincides with the span of Fourier-mode indicators $M_q^{\text{temp}} = \text{diag}(\delta_{k, q})$, and we use whichever basis is more convenient.

By Paper 44 § 3, $\mathcal{O}_{n_{\text{max}}, N_t, T}^L$ is a $*$ -closed linear subspace of $\mathcal{B}(\mathcal{K}_{n_{\text{max}}, N_t})$, contains the identity, and is *not* multiplicatively closed (a witness pair is constructed in Paper 44 § 5). The propagation number is $\text{prop}(\mathcal{O}_{n_{\text{max}}, N_t}^L) = 2$ on the Weyl-doubled achievable envelope, matching the Riemannian Paper 32 § III $\text{prop}=2$ verbatim; $\text{prop} = \infty$ on the full $\mathcal{B}(\mathcal{K})$ envelope due to the chirality-doubling and the commutative temporal subalgebra.

2.5 Joint central spectral Fejér kernel

The $\text{SU}(2)$ factor is the Paper 38 [36] L2-DEF central spectral Fejér kernel

$$K_{n_{\text{max}}}^{\text{SU}(2)}(g) := \frac{1}{Z_{n_{\text{max}}}^{\text{SU}(2)}} \left| \sum_{j \leq j_{\text{max}}} \sqrt{2j+1} \chi_j(g) \right|^2, \quad (9)$$

with $j_{\max} = (n_{\max} - 1)/2$ and $Z_{n_{\max}}^{\text{SU}(2)} = n_{\max}(n_{\max} + 1)/2$, central Plancherel symbol

$$\hat{K}_{n_{\max}}^{\text{SU}(2)}(j) = \frac{2j+1}{Z_{n_{\max}}^{\text{SU}(2)}} \mathbf{1}_{\{j \leq j_{\max}\}}. \quad (10)$$

The $U(1)$ factor is the standard Fejér kernel on S_T^1 ,

$$K_{N_t, T}^{U(1)}(\theta) := \frac{1}{N_t} \left| \sum_{|k| \leq K_{\max}} e^{i 2\pi k \theta / T} \right|^2, \quad (11)$$

with Plancherel symbol

$$\hat{K}_{N_t}^{U(1)}(k) = \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right). \quad (12)$$

The joint kernel is the tensor product

$$K_{n_{\max}, N_t, T}^{\text{joint}}(g, \theta) := K_{n_{\max}}^{\text{SU}(2)}(g) \cdot K_{N_t, T}^{U(1)}(\theta), \quad (13)$$

with factorized Plancherel symbol

$$K^{\text{joint}}(j, k) = \hat{K}_{n_{\max}}^{\text{SU}(2)}(j) \cdot \hat{K}_{N_t}^{U(1)}(k). \quad (14)$$

Both factors are non-negative, normalized (L^1 norm = 1), and central (class functions on their respective groups); the tensor product inherits these properties on $\text{SU}(2) \times U(1)$.

2.6 Krein-positive subspace and weak-form propinquity

The Krein-positive subspace is

$$\mathcal{K}^+ := \{|\psi\rangle \in \mathcal{K} : J_L|\psi\rangle = +|\psi\rangle\}, \quad (15)$$

the $+1$ -eigenspace of J_L , of dimension $\dim \mathcal{K}^+ = \dim \mathcal{K}/2 = \frac{1}{3} n_{\max}(n_{\max} + 1)(n_{\max} + 2)N_t$. The Krein inner product $\langle \psi, \phi \rangle_{J_L} := \langle \psi, J_L \phi \rangle$ restricted to $\mathcal{K}^+$ reduces to the standard Hilbert inner product $\langle \psi, \phi \rangle$, since $J_L|\phi\rangle = |\phi\rangle$ on $\mathcal{K}^+$. Hence $\mathcal{K}^+$ is a genuine Hilbert space.

The orthogonal projection onto $\mathcal{K}^+$ is $P_+ := (I + J_L)/2$ (and dually $P_- := (I - J_L)/2$ for the -1 -eigenspace).

Definition 2.1 ($\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity). For two Lorentzian truncated Krein metric spectral triples $\mathcal{T}_i^L = (\mathcal{O}_i^L, \mathcal{K}_i, D_i^L, J_i^L)$ with the construction above, the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity is

$$\Lambda^L(\mathcal{T}_1^L, \mathcal{T}_2^L) := \Lambda(\mathcal{T}_1^+, \mathcal{T}_2^+), \quad (16)$$

where $\mathcal{T}_i^+ := (P_i^+ \mathcal{O}_i^L P_i^+, \mathcal{K}_i^+, P_i^+ D_i^L P_i^+)$ is the $\mathcal{K}^+$ -restricted standard metric spectral triple and Λ is the standard Latrémolière metric-spectral-triple propinquity [16].

The construction is well-defined precisely because every Berezin-image operator and every truncation projection used below commutes with J_L at the operator level (Paper 44 § 4 / Sub-sprint C § 6 of the internal memos), so $\mathcal{K}^+$ is preserved and the standard machinery applies.

2.7 Joint gradient norms

For a pure-tensor symbol $f = f_s \otimes f_t$ on $S^3 \times S_T^1$, the L^1 -additive and L^2 -Pythagorean joint gradient norms are

$$\|\nabla^{\text{joint}, L^1} f\|_{\infty} := \|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} + \|f_s\|_{\infty} \cdot \|\partial_t f_t\|_{\infty}, \quad (17)$$

$$\|\nabla^{\text{joint}, L^2} f\|_{\infty} := \sup_{(x, t)} \sqrt{|\nabla_x f(x, t)|^2 + |\partial_t f(x, t)|^2}. \quad (18)$$

For general f , both extend by linearity over the Peter–Weyl $\times$ Fourier expansion and obey the standard triangle inequalities. The L^2 form is dominated by the L^1 form via $\sqrt{a^2 + b^2} \leq a + b$ for $a, b \geq 0$; both forms upper-bound the same Lipschitz seminorm with the same constant in the proof of Lemma 4.4.

3 Lemma L1': the operator-system substrate

Lemma L1' supplies the truncated metric spectral triple on which the four subsequent lemmas operate. It is the closure result of Paper 44 [41], summarized here for completeness.

Lemma 3.1 (L1', Paper 44). *The collection $\mathcal{T}_{n_{\max}, N_t, T}^L := (\mathcal{O}_{n_{\max}, N_t, T}^L, \mathcal{K}_{n_{\max}, N_t}, D_{n_{\max}, N_t, T}^L, J_{n_{\max}, N_t}^L)$ defined in § 2.2–2.4 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [32] / BBB [3]:*

- (a) $\mathcal{O}^L$ is a unital $*$ -closed linear subspace of $\mathcal{B}(\mathcal{K})$.
- (b) D_L is Krein-self-adjoint, $D_L^\times = D_L$.
- (c) J_L is a fundamental symmetry, $J_L^2 = +I$, $J_L^\dagger = J_L$.
- (d) The propagation number $\text{prop}(\mathcal{O}^L) = 2$ on the Weyl-doubled achievable envelope.
- (e) Every multiplier in $\mathcal{O}^L$ commutes with J_L : $[J_L, a] = 0$ bit-exact for every $a \in \mathcal{O}^L$.
- (f) At $N_t = 1$, the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on $\mathcal{H}_{\text{GV}}^{n_{\max}}$ (Frobenius residual 0.0 in float64).

Proof. (a)–(c) are direct from the construction and Paper 43 § 2,3,5. (d) is Paper 44 § 3 (via the structural identity $\dim((\mathcal{O}^L)^2) = \dim_{\text{Weyl}}^2 N_t$ on the Weyl-doubled envelope). (e) follows from $[\gamma^0, M^{\text{spat}}] = 0$ (chirality-swap acts trivially on the diagonal block-doubled spatial multiplier) and $[I_{N_t}, M^{\text{temp}}] = 0$ (trivially). (f) is Paper 44 § 2 (load-bearing falsifier F1). $\square$

Remark 3.2 (Krein-positivity at the operator-multiplier level is trivial). By Lemma 3.1(e), every operator $a \in \mathcal{O}^L$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$. Consequently, the $\mathcal{K}^+$ -restricted operator system $\mathcal{O}^{L,+} := P_+ \mathcal{O}^L P_+$ has the same propagation number 2 as the unrestricted system, and the standard Latrémolière metric-spectral-triple machinery applies on $(\mathcal{O}^{L,+}, \mathcal{K}^+, P_+ D_L P_+)$ verbatim. The non-trivial Krein-positive structure shifts to the *state* level (Wasserstein–Kantorovich on $\mathcal{K}^+$ states), which is the natural setting for the L5 propinquity assembly in § 5.

4 The three analytical lemmas

We now prove the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3) on the substrate supplied by Lemma 3.1. The mechanisms transfer from Paper 38 / Paper 40 with one substantive simplification per lemma, arising from the pure-tensor structure of $\mathcal{O}^L$ and the momentum-diagonal structure of the temporal multipliers.

4.1 Lemma L2: joint cb-norm of the central Schur multiplier

Lemma 4.1 (L2, joint cb-norm). *For all $n_{\max} \geq 1$ and $N_t \geq 1$, the central Schur multiplier $S_{K^{\text{joint}}}$ on the central subalgebra $Z(C(\text{SU}(2) \times U(1)))$ has cb-norm*

$$\|S_{K^{\text{joint}}}_{n_{\max}, N_t, T}\|_{\text{cb}} = \|S_{K^{\text{SU}(2)}}\|_{\text{cb}} \cdot \|S_{K^{U(1)}}\|_{\text{cb}} = \frac{2}{n_{\max} + 1} \cdot 1 = \frac{2}{n_{\max} + 1}. \quad (19)$$

Proof. The compact group $G = \text{SU}(2) \times U(1)$ is amenable (direct product of two compact groups, both amenable by Haar invariance [1, 25]). By the Bożejko–Fendler central-multiplier equality [4, 26] Ch. 8 Thm. 8.10, every central Fourier multiplier symbol $m \in \ell^\infty(\hat{G})$ has Schur cb-norm equal to its ℓ^∞ norm.

Factor 1 ($\text{SU}(2)$). Inherited from Paper 38 [36] Lemma L2(c) verbatim: the Plancherel symbol $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j) = (2j+1)/Z_{n_{\max}}^{\text{SU}(2)}$ is monotonically increasing in $j \in \{0, \frac{1}{2}, 1, \dots, j_{\max}\}$, so its supremum is $\hat{K}_{n_{\max}}^{\text{SU}(2)}(j_{\max}) = n_{\max}/Z_{n_{\max}}^{\text{SU}(2)} = 2/(n_{\max}+1)$.

Factor 2 ($U(1)$). $U(1)$ is abelian, so all irreps are 1-dimensional and every multiplier symbol is automatically central. The Fejér symbol (12) is non-negative, supported on $|k| \leq K_{\max}$, and monotonically decreasing in $|k|$ from $\hat{K}_{N_t}^{U(1)}(0) = 1$, so its supremum is 1, attained at $k = 0$.

Joint cb-norm via factorization. The central subalgebra of $C(\text{SU}(2) \times U(1))$ factorizes as the spatial C^* -tensor product $Z(C(\text{SU}(2))) \hat{\otimes} Z(C(U(1)))$ (Takesaki [29] Thm. IV.4.14, commutative-factor case). The joint Plancherel symbol (14) factorizes correspondingly. Both factor symbols are non-negative, so

$$\sup_{(j,k)} \hat{K}^{\text{SU}(2)}(j) \hat{K}^{U(1)}(k) = \left(\sup_j \hat{K}^{\text{SU}(2)}(j) \right) \left(\sup_k \hat{K}^{U(1)}(k) \right).$$

Applying Bożejko–Fendler in reverse to each factor (and once jointly on the amenable product) gives (19). $\square$

Remark 4.2 (N_t -independence of the cb-norm). The joint cb-norm $2/(n_{\max}+1)$ is independent of N_t : the $U(1)$ factor caps at 1 regardless of N_t . The N_t -dependence of the joint convergence rate enters the *mass-concentration moment* γ^{joint} in Lemma 4.9, not the cb-norm. This is the structurally correct behavior: the cb-norm controls the Lipschitz-distortion height in the propinquity assembly (§ 5), while γ^{joint} controls the reach. They are different invariants of the same kernel.

Remark 4.3 ($4/\pi$ asymptote on the $\text{SU}(2)$ factor). Paper 38 § 3.2 (with the universal Plancherel- $\times$ -Vandermonde cancellation theorem from Paper 40 § 3.2 [38]) gives the asymptotic rate constant

$$\lim_{n_{\max} \rightarrow \infty} \frac{n_{\max} \cdot \gamma_{n_{\max}}^{\text{SU}(2)}}{\log n_{\max}} = \frac{4}{\pi}.$$

This is the universal compact-Lie-group rate constant on the bi-invariant metric and is independent of rank, Weyl-group order, and Lie type within the class. The $U(1)$ factor inherits a separate Fejér-on-the-circle rate.

4.2 Lemma L3: joint Lichnerowicz / Lipschitz comparison

Lemma 4.4 (L3, joint Lichnerowicz). *For every $a \in \mathcal{O}_{n_{\max}, N_t, T}^L$ and every $(n_{\max}, N_t, T)$,*

$$\|[D_L, a]\|_{\text{op}} \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \|\nabla^{\text{joint}} a\|_{\infty}, \quad (20)$$

where ∇^{joint} is either the L^1 -additive or L^2 -Pythagorean joint gradient norm (17), (18). The constant satisfies

$$C_3^{\text{joint}}(n_{\max}, N_t) \leq C_3^{\text{SU}(2)}(n_{\max}) \leq \sup_{2 \leq N \leq 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} = \sqrt{1 - \frac{1}{n_{\max}}} \xrightarrow{n_{\max} \rightarrow \infty} 1^-, \quad (21)$$

where the supremum is taken over the natural-substrate envelope $N \leq 2n_{\max}-1$ — the Avery–Wen–Avery $\Delta n = \pm 1$ ladder on $n_{\max}$ shells produces shell-difference labels N up to $2n_{\max}-1$ via the chirality-doubling at the operator-system level (see Paper 44 [41] § 3). The per-harmonic function $(N-1)/\sqrt{N^2-1}$ is Paper 38 [36] Lemma L3 verbatim; the envelope range $N \leq 2n_{\max}-1$ extends Paper 38’s $N \leq n_{\max}$ because Paper 38’s Berezin map restricts to multiplier labels $N \leq n_{\max}$ by Plancherel cutoff, while the Lorentzian natural substrate’s full multiplier algebra reaches the larger envelope (see Remark 4.5 below).

Remark 4.5 (On the supremum range). The supremum is taken over the natural-substrate envelope $N \leq 2n_{\max} - 1$ rather than $N \leq n_{\max}$. The per-harmonic functional form, the asymptotic limit 1^- , and the main theorem (Theorem 5.3) are insensitive to this distinction. The finite-cutoff constant is sharpened by the envelope: at $n_{\max} = 3$, the natural-substrate generator $(N, L, M, p) = (4, 1, \pm 1, 0)$ has Lipschitz ratio 0.95, which exceeds the per-shell value $C_3^{(3)} = 2/\sqrt{8} = 0.707$ but is bounded by the envelope-aware $C_3^{\text{op}}(3) = \sqrt{(2 \cdot 3 - 1) - 1}/\sqrt{(2 \cdot 3 - 1)^2 - 1} = \sqrt{2/3} \approx 0.816$. Numerical-panel values in §6.1 are unaffected (they quote γ^{joint} directly, not $C_3^{\text{joint}} \cdot \gamma^{\text{joint}}$). The envelope refinement was identified in Paper 46 [42] § 4.3 (the strong-form companion under $L_{\text{op}}(a) := \|[D_L, a]\|_{\text{op}}$); Paper 38 is unaffected because Paper 38’s Berezin map restricts multiplier labels to $N \leq n_{\max}$ by Plancherel cutoff, while the Lorentzian natural substrate’s chirality-doubled multiplier algebra reaches the larger envelope (see Paper 46 Appendix A for the cross-paper analysis).

Proof. We first establish the structural identity: for every pure-tensor $a = a_s \otimes a_t \in \mathcal{O}^L$,

$$[D_L, a_s \otimes a_t] = i [D_{\text{GV}}, a_s] \otimes a_t \quad (\text{bit-exact}). \quad (22)$$

Expand $[D_L, a_s \otimes a_t] = i [\gamma^0 \otimes \partial_t, a_s \otimes a_t] + i [D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t]$ and analyze each term.

Term A (vanishing cross term). Since $a_s = \text{blkdiag}(W, W)$ has identical diagonal blocks in chirality and γ^0 is the chirality-swap acting as $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$ on the chirality indices,

$$\gamma^0 a_s = \begin{pmatrix} 0 & W \\ W & 0 \end{pmatrix} = a_s \gamma^0, \quad [\gamma^0, a_s] = 0.$$

Furthermore, a_t and ∂_t are both diagonal in the momentum basis (the temporal multiplier algebra is the commutative diagonal subalgebra of $M_{N_t}(\mathbb{C})$), so $[\partial_t, a_t] = 0$. The cross term therefore vanishes bit-exactly: $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$.

Term B (purely spatial). $[D_{\text{GV}} \otimes I_{N_t}, a_s \otimes a_t] = [D_{\text{GV}}, a_s] \otimes a_t$.

Combining gives (22). By linearity over the operator-system span, the same identity extends to every $a \in \mathcal{O}^L$ written as a finite linear combination of pure-tensor generators.

Tensor-product operator-norm factorization $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}} \cdot \|B\|_{\text{op}}$ and Paper 38 Lemma L3 on the spatial factor give, for any pure-tensor $a = a_s \otimes a_t$,

$$\|[D_L, a]\|_{\text{op}} = \|[D_{\text{GV}}, a_s]\|_{\text{op}} \cdot \|a_t\|_{\text{op}} \leq C_3^{\text{SU}(2)}(n_{\max}) \cdot \|\nabla_x f_s\|_{\infty} \cdot \|a_t\|_{\text{op}}, \quad (23)$$

where f_s is the Berezin pre-image of a_s . For momentum-diagonal a_t , $\|a_t\|_{\text{op}} = \|f_t\|_{\infty}$ (with f_t the corresponding Fourier symbol). Since $\|\nabla_x f_s\|_{\infty} \cdot \|f_t\|_{\infty} \leq \|\nabla^{\text{joint}, L^1} f\|_{\infty}$ by (17) (the second summand is non-negative), and similarly $\leq \|\nabla^{\text{joint}, L^2} f\|_{\infty}$ by $\sqrt{a^2 + b^2} \geq a$ for $a, b \geq 0$, we obtain (20) for pure-tensor a . The extension to general $a = \sum_j c_j (a_s^j \otimes a_t^j)$ follows by triangle inequality on both sides.

The bound (21) takes the Paper 38 § L3 per-harmonic form $(N-1)/\sqrt{N^2-1}$ on the natural-substrate envelope $N \leq 2n_{\max} - 1$ (per Remark 4.5); asymptotic tightness $\rightarrow 1^-$ is via the per-harmonic Avery–Wen–Avery saturation on $Y_{N,0,0}^{(3)}$ as $N \rightarrow 2n_{\max} - 1$. $\square$

Remark 4.6 (The temporal direction contributes zero). The identity (22) is the substantive structural content of Lemma 4.4. There is *no* cross term and *no* separate $U(1)$ Lichnerowicz lemma: the joint commutator is the spatial commutator times the temporal operator-norm, and the temporal slot enters the bound multiplicatively at value 1. This is stronger than the $C_3 \leq 1 + \text{cross estimate}$ one would expect from a generic tensor extension; the structural reason is that the temporal multiplier algebra is commutative diagonal and ∂_t is itself diagonal, so they commute bit-exactly. The Wick-rotated convention here is essential: a Lorentzian-causal temporal derivative would not be diagonal in the same basis.

Remark 4.7 (L^1 and L^2 give the same C_3). Both L^1 -additive (17) and L^2 -Pythagorean (18) joint gradient norms upper-bound the right-hand side of (23) by the same constant $C_3^{\text{SU}(2)}$: the temporal contribution to $\| [D_L, a] \|_{\text{op}}$ is zero (by (22)), so only the spatial part of the joint gradient matters in the inequality.

4.3 Lemma L4: joint Berezin reconstruction

Definition 4.8 (Joint Berezin reconstruction). The joint Berezin reconstruction map is $B_{n_{\max}, N_t, T}^{\text{joint}} : C^\infty(S^3 \times S_T^1) \rightarrow \mathcal{O}_{n_{\max}, N_t, T}^L$ defined on the joint Peter–Weyl $\times$ Fourier expansion $f(\omega, t) = \sum_{N, L, M, q} c_{N, L, M, q} Y_{N, L, M}^{(3)}(\omega) e^{iqt/R_T}$ (with $R_T = T/(2\pi)$) by

$$B^{\text{joint}}(f) = \sum_{\substack{N \leq n_{\max} \\ |q| \leq K_{\max}}} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \hat{K}_{n_{\max}}^{\text{SU}(2)}(N) \hat{K}_{N_t}^{U(1)}(q) c_{N, L, M, q} (M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}). \quad (24)$$

Equivalently, $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$ where $P^{\text{joint}} := P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$ is the joint truncation projection onto the truncated subspace of $L^2(S^3 \times S_T^1)$ spanned by the kept Peter–Weyl $\times$ Fourier modes. The two definitions agree on the central subalgebra by joint Plancherel (14).

Lemma 4.9 (L4, joint Berezin properties). $B_{n_{\max}, N_t, T}^{\text{joint}}$ satisfies the following four properties.

- (a) Positivity. If $f \geq 0$ pointwise on $S^3 \times S_T^1$, then $B^{\text{joint}}(f) \geq 0$ as a positive-semidefinite Hermitian element of $M_{\dim \mathcal{K}}(\mathbb{C})$.
- (b) Contractivity. $\|B^{\text{joint}}\|_{\text{cb}} \leq \|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1) \leq 1$, and in particular $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|f\|_\infty$.
- (c) Approximate identity. For every $f \in C^\infty(S^3 \times S_T^1)$,

$$\|B^{\text{joint}}(f) - P^{\text{joint}} M_f P^{\text{joint}}\|_{\text{op}} \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_\infty, \quad (25)$$

with $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \rightarrow 0$ as $(n_{\max}, N_t) \rightarrow (\infty, \infty)$.

- (d) L3 compatibility. For every $f \in C^\infty(S^3 \times S_T^1)$,

$$\|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \leq C_3^{\text{joint}} \cdot \|\nabla^{\text{joint}} f\|_\infty, \quad (26)$$

with the same $C_3^{\text{joint}} \leq 1$ as in Lemma 4.4.

- (e) Krein-positivity preservation. For every f , $[J_L, B^{\text{joint}}(f)] = 0$ bit-exact, hence $B^{\text{joint}}(f)$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$.

Proof. (a) *Positivity.* The convolution form $B^{\text{joint}}(f) = P^{\text{joint}} (K^{\text{joint}} * f) P^{\text{joint}}$ writes $B^{\text{joint}}(f)$ as a UCP compression of a non-negative function (since both factor kernels are non-negative pointwise: $K^{\text{SU}(2)} = |\cdot|^2 \cdot Z^{-1} \geq 0$ and $K^{U(1)} = |\cdot|^2 \cdot N_t^{-1} \geq 0$). The convolution of two non-negative integrands against the positive joint Haar measure is non-negative, and the UCP compression of a non-negative multiplication operator is positive-semidefinite.

(b) *Contractivity.* The convolution form gives $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_\infty \leq \|K^{\text{joint}}\|_{L^1} \cdot \|f\|_\infty$ by Young’s inequality at the sup endpoint, and $\|K^{\text{joint}}\|_{L^1} = 1$ (both factors are normalized probability densities; Haar on a product factorizes). The sharper cb-norm bound is via Lemma 4.1: B^{joint} is the composition of the UCP projection P^{joint} (cb-norm 1) with the central Schur multiplier $S_{K^{\text{joint}}}$ (cb-norm $2/(n_{\max}+1)$ by Lemma 4.1), giving $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max}+1)$.

(c) *Approximate identity.* For a pure-tensor symbol $f = f_s \otimes f_t$, the convolution-deficit splits:

$$K^{\text{joint}} * f - f = (K^{\text{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t).$$

By Paper 38 [36] Lemma L4(c) on the spatial factor, $\|K^{\text{SU}(2)} * f_s - f_s\|_\infty \leq \gamma_{n_{\max}}^{\text{SU}(2)} \cdot \|\nabla_x f_s\|_\infty$, with $\gamma_{n_{\max}}^{\text{SU}(2)} = O(\log n_{\max}/n_{\max})$ (Paper 38 § L2 quantitative rate, $4/\pi$ asymptote). By the standard Fejér-on-the-circle approximation rate (Katznelson [15] Ch. I), $\|K^{U(1)} * f_t - f_t\|_\infty \leq \gamma_{N_t, T}^{U(1)} \cdot \|\partial_t f_t\|_\infty$ with $\gamma^{U(1)} = O(T/N_t)$. Combining with the L^1 -additive triangle decomposition and $\|K^{U(1)} * f_t\|_\infty \leq \|f_t\|_\infty$ (Young) gives

$$\|K^{\text{joint}} * f - f\|_\infty \leq \max(\gamma_{n_{\max}}^{\text{SU}(2)}, \gamma_{N_t, T}^{U(1)}) \cdot \|\nabla^{\text{joint}, L^1} f\|_\infty.$$

We define $\gamma^{\text{joint}} := \max(\gamma^{\text{SU}(2)}, \gamma^{U(1)})$. Compression-contraction on L^2 does not increase the operator norm, so (25) follows. Linearity and the triangle inequality extend to general f .

(d) *L3 compatibility.* By linearity over (24), $[D_L, B^{\text{joint}}(f)] = \sum c_{N, L, M, q} K^{\hat{\text{joint}}}(N, q) i[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}] \otimes M_q^{\text{temp}}$ by the L3 structural identity (22). Tensor-norm factorization and Paper 38 § L3 give $\|[D_{\text{GV}}, M_{N, L, M}^{\text{spat}}]\|_{\text{op}} \leq (N-1) \cdot \|M^{\text{spat}}\|_{\text{op}}$ with $C_3^{(N)} = (N-1)/\sqrt{N^2-1}$ on the natural Avery harmonic. Combining with $K^{\hat{\text{joint}}}(N, q) \leq 1$ and triangle inequality gives (26).

(e) *Krein-positivity preservation.* $[J_L, M_{N, L, M}^{\text{spat}} \otimes M_q^{\text{temp}}] = [J_L^{\text{spat}}, M^{\text{spat}}] \otimes M^{\text{temp}} + M^{\text{spat}} \otimes [I_{N_t}, M^{\text{temp}}] = 0$ by the same argument as Lemma 3.1(e), extended by linearity. $\square$

Remark 4.10 (Tensor-product factorization on pure tensors). For a pure-tensor $f = f_s \otimes f_t$, the joint Berezin factorizes:

$$B^{\text{joint}}(f_s \otimes f_t) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes B^{U(1)}(f_t), \quad (27)$$

where $B_{\chi\text{-d}}^{\text{SU}(2)}$ is the chirality-doubled spinor lift of Paper 38's spatial Berezin and $B^{U(1)}$ is the standard Fejér-Berezin on S_T^1 . This factorization makes Lemma 4.9(a)–(e) follow from the corresponding factor-wise properties of Paper 38 L4 (SU(2) side) and standard Fejér-on- S^1 approximation theory ($U(1)$ side).

5 Main theorem (Lemma L5): propinquity assembly

5.1 The tunneling pair

Definition 5.1 (Joint tunneling pair). The joint tunneling pair between the truncated triple $\mathcal{T}_{n_{\max}, N_t, T}^L$ and the continuum reference triple $\mathcal{T}_{S^3 \times S_T^1}^L$ is the pair $(B_{n_{\max}, N_t, T}^{\text{joint}}, P_{n_{\max}, N_t}^{\text{joint}})$, with B^{joint} the joint Berezin map of Definition 4.8 and $P^{\text{joint}} = P_{n_{\max}}^{\text{SU}(2)} \otimes P_{N_t}^{U(1)}$ the joint truncation projection on $L^2(S^3 \times S_T^1)$.

The continuum reference triple is $\mathcal{T}_{S^3 \times S_T^1}^L := (C^\infty(S^3 \times S_T^1), L^2(S^3, \Sigma_{\text{CH}}) \otimes L^2(S_T^1), D_\infty^L, J_\infty^L)$ with Σ_{CH} the chirality-doubled Camporesi–Higuchi spinor bundle and D_∞^L, J_∞^L the natural continuum analogs of D_L and J_L .

By Lemmas 3.1–4.9, both B^{joint} and P^{joint} are UCP (with the Berezin cb-norm bound $\|B^{\text{joint}}\|_{\text{cb}} \leq 2/(n_{\max} + 1) \leq 1$; the projection P^{joint} is a Stinespring orthogonal projection), and both commute with J_L at the operator level. Hence the $\mathcal{K}^+$ -restricted pair $(P_+ B^{\text{joint}}(\cdot) P_+, P_+ P^{\text{joint}}(\cdot) P_+)$ is a valid Latrémolière tunneling pair between the $\mathcal{K}^+$ -restricted truncated and continuum triples in Definition 2.1.

5.2 The four propinquity constituents

Following Latrémolière [16]§ 4 (metric-spectral-triple weak-form propinquity bound) and Paper 38 § L5, the propinquity bound for a direct UCP tunneling pair is

$$\Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \leq \max(\text{reach}_B, \text{reach}_P, \text{height}_B, \text{height}_P). \quad (28)$$

Proposition 5.2 (Reach and height bounds). *The four propinquity constituents satisfy:*

$$\text{reach}_B \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (29)$$

$$\text{reach}_P \leq \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (30)$$

$$\text{height}_B \leq C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}, \quad (31)$$

$$\text{height}_P = 0. \quad (32)$$

Proof. (29) is Lemma 4.9(c) (approximate identity), applied on the unit-Lipschitz ball $\{f : \|\nabla^{\text{joint}} f\|_\infty \leq 1\}$. The bound restricts unchanged to $\mathcal{K}^+$ since P_+ is UCP.

(30) is the dual roundtrip: start with M_f on the continuum side, apply P^{joint} to compress to $\mathcal{O}^L$, and lift back with a bounded partial inverse σ of B^{joint} on the central subalgebra. By Lemma 4.1, $\|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1)$ is finite and non-zero, so σ exists on the central subalgebra and is bounded by the same scale; the dual residual is symmetric to (29).

(31) is the Lipschitz-distortion height of Latrémolière [16] Def. 3.5, given by

$$\text{height}_B = \sup_{\|\nabla^{\text{joint}} f\|_\infty \leq 1} \left| \|f\|_{\text{Lip}} - \|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} \right|,$$

where $\|B^{\text{joint}}(f)\|_{\text{Lip}^{\text{op}}} := \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}}$ is the Lipschitz seminorm induced by the truncated Lorentzian Dirac. By Lemma 4.9(d) and an L4 approximate-identity step on the gradient (Stein-Weiss / Paper 38 Appendix A factor-wise), the height is bounded by $C_3^{\text{joint}} \cdot \gamma^{\text{joint}}$.

(32) follows because P^{joint} is an orthogonal projection (Stinespring lift), of cb-norm 1, and introduces no Lipschitz distortion. $\square$

5.3 Main theorem

Theorem 5.3 (Lorentzian propinquity convergence on truncated $\text{SU}(2) \times U(1)_T$ Krein spectral triples). *Let $n_{\max} \geq 1$, $N_t \geq 1$, $T > 0$. The $\mathcal{K}^+$ -restricted weak-form Lorentzian Latrémolière propinquity between the truncated Krein metric spectral triple $\mathcal{T}_{n_{\max}, N_t, T}^L$ and the continuum reference triple $\mathcal{T}_{S^3 \times S_T^1}^L$ satisfies*

$$\Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) \leq C_3^{\text{joint}}(n_{\max}, N_t) \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}} \xrightarrow{(n_{\max}, N_t) \rightarrow (\infty, \infty)} 0, \quad (33)$$

with $C_3^{\text{joint}}(n_{\max}, N_t) \leq 1$ asymptotically tight (inherited from Paper 38 / Lemma 4.4) and $\gamma_{n_{\max}, N_t, T}^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ the joint mass-concentration moment (Lemma 4.9(c)).

Proof. By Definition 2.1 and Proposition 5.2,

$$\begin{aligned} \Lambda^L(\mathcal{T}_{n_{\max}, N_t, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) &= \Lambda(\mathcal{T}_{n_{\max}, N_t, T}^+, \mathcal{T}_{S^3 \times S_T^1}^+) \\ &\leq \max(\gamma^{\text{joint}}, \gamma^{\text{joint}}, C_3^{\text{joint}} \cdot \gamma^{\text{joint}}, 0) \\ &= C_3^{\text{joint}} \cdot \gamma_{n_{\max}, N_t, T}^{\text{joint}}. \end{aligned}$$

Since $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ vanishes in the joint limit $(n_{\max}, N_t) \rightarrow (\infty, \infty)$ and $C_3^{\text{joint}} \leq 1$ asymptotically, the right-hand side $\rightarrow 0$. $\square$

Remark 5.4 (Asymptotic rate). The joint rate factors at leading order:

$$\gamma^{\text{joint}} \leq \gamma^{\text{SU}(2)} + \gamma^{U(1)} = O\left(\frac{\log n_{\max}}{n_{\max}}\right) + O\left(\frac{T}{N_t}\right),$$

with the $\text{SU}(2)$ leading constant $\lim_{n_{\max} \rightarrow \infty} n_{\max} \gamma^{\text{SU}(2)} / \log n_{\max} = 4/\pi$ rigorous (Paper 38 § L2 Stein–Weiss closure + Paper 40 § 3.2 universal cancellation), and the $U(1)$ leading constant $\gamma_{N_t, T}^{U(1)} \leq C T \log N_t / N_t$ standard (Katznelson [15] Ch. I).

Remark 5.5 (Limit identification). The propinquity limit point is the $\mathcal{K}^+$ -restricted standard metric spectral triple $\mathcal{T}_{S^3 \times S_T^1}^+ = (P_+ C^\infty(S^3 \times S_T^1) P_+, \mathcal{K}_\infty^+, P_+ D_\infty^L P_+)$ on the $\mathcal{K}^+$ subspace of the continuum Krein space. By Latrémolière’s weak-form theorem [16] Thm. 5.5, the limit identification of Λ -convergence is on the Wasserstein–Kantorovich state space of the limit triple; this carries over to Λ^L via Definition 2.1. The weak-form caveat is that the strong-form Lorentzian propinquity (without $\mathcal{K}^+$ restriction) would identify a Krein-state-space limit, which remains open; see § 7.3.

Remark 5.6 (The proof shape is bookkeeping). In the same spirit as Paper 38 § L5 and Leimbach–vS [18], the propinquity assembly is the bookkeeping layer of the proof shape. The analytical content is supplied by Lemmas 3.1, 4.1, 4.4, 4.9; each step in Proposition 5.2 cites one of them. No new analytical input enters the assembly.

Remark 5.7 (Two-rate hybrid convergence to the non-compact carrier). The construction here is at *fixed* compact temporal radius T (canonical $T = 2\pi$). At any *admissible scaling* $T(N_t) \rightarrow \infty$ with $T(N_t)/N_t \rightarrow 0$ as $N_t \rightarrow \infty$, the present theorem applies at every coupled cell $(n_{\max}, N_t, T(N_t))$ with $C_3^{\text{joint}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ (T -independent by L3 structural identity (22)) and $\gamma^{\text{joint}} = \max(O(\log n_{\max}/n_{\max}), O(T(N_t)/N_t)) \rightarrow 0$ (the $U(1)_T$ factor of L4 carries the only T -dependence in the bound, and it is the standard Fejér-on-circle rate). Composed with the norm-resolvent convergence $\mathcal{T}_{S^3 \times S_T^1}^L \rightarrow \mathcal{T}_{S^3 \times \mathbb{R}_t}^L$ of Paper 47 [43] § 4, this gives a *two-rate hybrid convergence* statement

$$\mathcal{T}_{n_{\max}, N_t, T(N_t)}^L \xrightarrow{\Lambda^L\text{-rate } O(\log n_{\max}/n_{\max} + T(N_t)/N_t)} \mathcal{T}_{S^3 \times S_{T(N_t)}^1}^L \xrightarrow{\text{nr-rate } O(e^{-|\text{Im } z| T(N_t)/2})} \mathcal{T}_{S^3 \times \mathbb{R}_t}^L$$

identifying the non-compact Lorentzian Dirac as the spectral limit of the framework’s compact-temporal sequence. The propinquity-level closure on the non-compact carrier (single uniform Latrémolière bound to the non-compact target) remains open and is named G2-metric (Paper 47 § 8 Q1).

6 Numerical verification and Riemannian-limit recovery

6.1 Verification panel

The driver `debug/13b_2_sub_sprint_D_compute.py` builds the joint tunneling pair on the panel

$$(n_{\max}, N_t) \in \{(2, 1), (3, 1), (4, 1), (2, 3), (3, 5), (4, 7)\}$$

at $T = 2\pi$. The panel verifies four assertions: (i) Riemannian-limit recovery at $N_t = 1$; (ii) monotone decrease of Λ^L in $n_{\max}$ at fixed N_t ; (iii) asymptotic rate consistency $\Lambda^L(4, 7)/\Lambda^L(2, 3) \approx 0.637$; (iv) constituent identities $\text{reach}_B, \text{reach}_P \leq \gamma^{\text{joint}}$ at every cell.

The numerical results are tabulated in Table 1 (reproduced from `debug/data/13b_2_sub_sprint_D.json`; see Appendix B for full data).

$(n_{\max}, N_t)$	$\gamma^{\text{SU}(2)}$	$\gamma^{U(1)}$	C_3^{joint}	cb-norm	Λ^L bound	$N_t = 1$ residual
(2, 1)	2.0746	$\pi/2 \approx 1.5708$	0.5774	2/3	2.0746	0.0 bit-exact
(3, 1)	1.6101	$\pi/2$	0.7071	1/2	1.6101	0.0 bit-exact
(4, 1)	1.3223	$\pi/2$	0.7746	2/5	1.3223	0.0 bit-exact
(2, 3)	2.0746	0.7220	0.5774	2/3	2.0746	—
(3, 5)	1.6101	0.4956	0.7071	1/2	1.6101	—
(4, 7)	1.3223	0.3841	0.7746	2/5	1.3223	—

Table 1: Numerical verification panel for Theorem 5.3. $\gamma^{\text{SU}(2)}$ values from Paper 38 § L2 closed-form Stein–Weiss computation; $\gamma^{U(1)}$ values from standard Fejér-on- S_T^1 at $T = 2\pi$; C_3^{joint} from the Avery harmonic saturation $(N - 1)/\sqrt{N^2 - 1}$ on the truncation. The bound on Λ^L is dominated by the SU(2) factor at every cell. The $N_t = 1$ Riemannian-limit residuals are bit-exact 0.0 in float64, the load-bearing falsifier for the construction.

6.2 Riemannian-limit recovery (load-bearing falsifier)

Proposition 6.1 (Riemannian-limit recovery). *At $N_t = 1$, the $\mathcal{K}^+$ -restricted weak-form Lorentzian propinquity reduces bit-exactly to Paper 38’s SU(2) Riemannian propinquity:*

$$\Lambda^L(\mathcal{T}_{n_{\max}, 1, T}^L, \mathcal{T}_{S^3 \times S_T^1}^L) = \Lambda_{n_{\max}}^{\text{Paper 38}} \quad (\text{Frobenius residual 0.0 in float64}). \quad (34)$$

Proof. At $N_t = 1$, the only kept Fourier mode is $q = 0$, the $U(1)$ Fejér kernel collapses to the uniform Haar density $1/T$, and the temporal Berezin factor $B^{U(1)}$ is the trivial 1×1 identity. By the tensor-product factorization (27), $B_{n_{\max}, 1, T}^{\text{joint}}(f_s \otimes \mathbf{1}) = B_{\chi\text{-d}}^{\text{SU}(2)}(f_s) \otimes I_1$, which equals Paper 38’s chirality-doubled spinor lift of the spatial Berezin map.

The chirality-doubled spinor lift on $\mathcal{K}^+$ is bit-identical to Paper 38’s scalar Fock-basis construction up to the basis representation: the Plancherel weights $\hat{K}^{\text{SU}(2)}(N)$, the Lipschitz bound constant $C_3^{\text{Paper 38}} \leq 1$, and the convergence rate $\gamma_{n_{\max}}^{\text{SU}(2)}$ are the same numerical objects. Therefore (34) holds. The bit-exact residual at $n_{\max} \in \{2, 3, 4\}$ is verified numerically in Table 1. $\square$

The Riemannian-limit recovery is the load-bearing falsifier of the construction: were it to fail, we would have introduced an extraneous factor in the Wick-rotated signature, miscounted Berezin weights or grids, or had a sign error in the $\mathcal{K}^+$ restriction. The bit-exact recovery rules out all three. This is the same load-bearing-falsifier pattern used in Paper 42 (Riemannian limit of the Tomita–Takesaki construction at $N_t = 1$ to Paper 38 / Paper 42 § 11) and Paper 43 (Riemannian limit of the Lorentzian Dirac at $N_t = 1$ to truthful Camporesi–Higuchi spatial Dirac).

6.3 Asymptotic rate consistency

The ratio $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$ is monotone decreasing across the panel, matching the single-factor Paper 38 and Paper 39 [37] ratios bit-identically. This confirms that the SU(2) factor dominates the joint rate at the tested cutoffs, which is the expected behavior given $\gamma^{\text{SU}(2)} = O(\log n_{\max}/n_{\max})$ and $\gamma^{U(1)} = O(T/N_t)$: for $N_t \gtrsim n_{\max}/\log n_{\max}$, the SU(2) factor controls the rate.

7 Discussion

7.1 Four-witness Wick-rotation connection

Paper 42 [39] closes the four-witness Wick-rotation theorem (Hartle–Hawking [12], Sewell [27], Bisognano–Wichmann [2], Unruh [31]) at the operator-system level on the *Riemannian* truncated Camporesi–Higuchi spectral triple, via the geometric BW- α and Tomita–Takesaki BW- γ modular Hamiltonians on a hemispheric wedge. Paper 43 [40] extends that closure to signature $(3, 1)$ on the Krein wedge.

The present paper supplies the Lorentzian *convergence* leg of the arc: Theorem 5.3 shows that the family of truncated Krein spectral triples on which Paper 42/43 work converges in the $\mathcal{K}^+$ -restricted weak-form propinquity to the continuum reference triple as $(n_{\max}, N_t) \rightarrow \infty$. The triple of papers (42, 43, 45) together constitute the Lorentzian-extension closure of the math.OA quartet (38, 39, 40 on the Riemannian side).

7.2 Relation to Mondino–Sämman

The synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämman [20] extends their earlier work [19] to Lorentzian pre-length spaces via ϵ -nets of causal diamonds. This is adjacent at the conceptual level (both “Lorentzian Gromov–Hausdorff-type convergence”) but operates on a categorically different mathematical object.

Remark 7.1 (Different objects, different distances). Mondino–Sämman work on pre-length spaces with causal-diamond geometry; they do not have a Dirac operator or any spectral data, and their convergence is in the synthetic-geometry metric on the space of isometry classes of such pre-length spaces. The present paper works on truncated Krein spectral triples; the convergence is in the operator-algebraic Latrémolière propinquity, restricted to the Krein-positive state-space substrate of the operator system. Neither construction implies the other; both are valid notions of Lorentzian-type Gromov–Hausdorff convergence on their respective categories.

7.3 Open questions

- (Q1) **Strong-form Lorentzian propinquity.** A Latrémolière-style metric on Krein-signature spectral triples *without* the $\mathcal{K}^+$ restriction remains an open NCG-math problem. The technical issue: the standard Latrémolière construction uses the Lipschitz seminorm $\|[D, \cdot]\|_{\text{op}}$, which is well-defined on Hilbert space but requires an indefinite-inner-product analog on a Krein space with appropriate Krein-self-adjointness on D_L per vdD / BBB. The published math.OA literature as of May 2026 does not provide this construction. By performing the $\mathcal{K}^+$ restriction we trade the strong-form construction for one that uses Hilbert-space machinery on $\mathcal{K}^+$ and obtains a defensible weak-form convergence theorem at the $\mathcal{K}^+$ -state-space / Wasserstein–Kantorovich level. Extending to strong form is a multi-month original NCG-math project, and the structural sharpening of Paper 42 § 10 question O3 (the spectral-action-vs-modular-Hamiltonian distinction) suggests the strong form will need additional structural input.
- (Q2) **De-compactification limit $T \rightarrow \infty$.** The present construction is at compact temporal radius T . The $T \rightarrow \infty$ limit recovers the non-compact $\mathbb{R}_t$ direction; the joint mass-concentration moment $\gamma^{U(1)} = O(T/N_t)$ then requires N_t to grow at least as fast as T , and the asymptotic rate analysis must be re-examined in this regime. This is internally a separate sprint (L3c in our taxonomy) and is not addressed here.
- (Q3) **Cross-manifold extensions.** The cross-manifold extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (Paper 24 [34] § V). The four-layer Coulomb/HO asymmetry there (spectrum-computing L_0 , calibration π ,

Wilson gauge with natural matter, modular-Hamiltonian structure of the wedge KMS state) precludes the natural extension within the present framework.

- (Q4) **Sharp asymptotic-tight** $C_3^{\text{joint}} = 1$. The present bound has $C_3^{\text{joint}} \leq 1$ asymptotically tight, with finite-cutoff bound $\sup_{N \leq 2n_{\max}-1} (N-1)/\sqrt{N^2-1} = \sqrt{1-1/n_{\max}} < 1$ (per Remark 4.5). A direct proof of $C_3^{\text{joint}} = 1$ in the joint limit (rather than inheriting from the $\text{SU}(2)$ factor) would require a finer joint analysis of the temporal direction's contribution to the Lipschitz seminorm when restricted to non-pure-tensor multipliers; this is not needed for the qualitative-rate convergence theorem here.
- (Q5) **Quantitative rate constants on the $U(1)$ factor.** The $U(1)$ leading constant in $\gamma^{U(1)} = O(T \log N_t/N_t)$ is standard Cesàro/Fejér; a Stein–Weiss-style derivation of the joint constant matching the $\text{SU}(2)$ factor's $4/\pi$ would be a useful refinement, but is not needed for the qualitative-rate convergence theorem.

7.4 Structural-skeleton scope

The construction obeys the structural-skeleton scope discipline of the GeoVac framework: the convergence theorem determines the rate, the asymptotic constant, and the limit identification on the $\mathcal{K}^+$ -substrate at finite cutoff. It does not determine calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of $T = 2\pi$ is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any $T > 0$.

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A Symbolic Plancherel calculation for the joint Fejér symbol

We record the symbolic factorization of the joint Plancherel symbol $K^{\text{joint}}(j, k)$ used in Lemma 4.1.

Lemma A.1. *For all $n_{\max} \geq 1$, $N_t \geq 1$, the joint Plancherel symbol factorizes exactly in $\mathbb{Q}$:*

$$K^{\text{joint}}_{n_{\max}, N_t}(j, k) = \hat{K}^{\text{SU}(2)}_{n_{\max}}(j) \cdot \hat{K}^{U(1)}_{N_t}(k),$$

with

$$\hat{K}^{\text{SU}(2)}_{n_{\max}}(j) = \frac{2j+1}{Z^{\text{SU}(2)}_{n_{\max}}} \mathbf{1}_{\{j \leq j_{\max}\}}, \quad Z^{\text{SU}(2)}_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2},$$

$$\hat{K}^{U(1)}_{N_t}(k) = \max\left(0, \frac{N_t+1-2|k|}{N_t+1}\right).$$

Proof. For the $\text{SU}(2)$ factor, the spectral Fejér kernel $K^{\text{SU}(2)} = Z^{-1} |\sum_{j \leq j_{\max}} \sqrt{2j+1} \chi_j|^2$ has Plancherel symbol computed by orthogonality of characters on $\text{SU}(2)$: $\int_{\text{SU}(2)} K^{\text{SU}(2)} \chi_{j'} dg = (2j'+1)/Z$ for $j' \leq j_{\max}$, zero otherwise (Paper 38 § L2(a)).

For the $U(1)$ factor, the Fejér kernel $K^{U(1)} = N_t^{-1} |D_{N_t}|^2$ where $D_{N_t} = \sum_{|k| \leq K_{\max}} e^{i 2\pi k \theta / T}$ is the Dirichlet kernel. Direct geometric-series computation gives $K^{\hat{U}(1)}(k) = \max(0, (N_t+1-2|k|)/(N_t+1))$ (Katznelson [15] Ch. I.2).

The joint kernel is the tensor product on the direct product group; the joint Plancherel symbol on the dual $\hat{\text{SU}}(2) \times \hat{U}(1)$ factorizes correspondingly via Peter–Weyl on $\text{SU}(2) \times U(1)$ (direct product gives external tensor products of irreps; characters multiply). $\square$

B Numerical verification panel data

The full numerical panel data tabulated in Table 1 is reproduced from `debug/data/l3b_2_sub_sprint_D.json`. The corresponding computational driver is `debug/l3b_2_sub_sprint_D_compute.py`, and the production module is `geovac/lorentzian_propinquity_compact_temporal.py` (approximately 700 lines, with the `LorentzianTunnelingPair` dataclass, the `LorentzianPropinquityBound` dataclass, the `compute_lorentzian_propinquity_bound` function, and the `verify_riemannian_limit_at_N_t` bit-exact check).

The numerical verification of all constituents in Proposition 5.2 is in `tests/test_lorentzian_propinquity.py`. (15+ test functions covering the tunneling pair construction, reach and height bounds, the main theorem panel, the asymptotic rate consistency, the $\mathcal{K}^+$ restriction correctness, the Riemannian-limit recovery, and the monotone decrease in $n_{\max}$).

Convergence diagnostic. The ratio $\Lambda^L(4, 7)/\Lambda^L(2, 3) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38 / Paper 39 ratio bit-identically, confirming that the $\text{SU}(2)$ factor dominates the joint rate at the tested cutoffs. At larger cutoffs the $U(1)$ factor would re-enter at order $O(T/N_t)$ if N_t does not grow as fast as $n_{\max}/\log n_{\max}$.

Memory note. The panel cell $(n_{\max}, N_t) = (5, 9)$ was originally planned but exceeds the operator-system vec-matrix memory budget at $\dim_K = 1260$ and $K = 2565$ multipliers (≈ 60 GiB `complex128`). The panel is restricted to $\{(2, 3), (3, 5), (4, 7)\}$, which is sufficient for Riemannian-limit recovery, monotone-decrease, and asymptotic-rate consistency. Larger cells would require sparse linear-algebra refactoring of `geovac.operator_system_compact_temporal`, which is a downstream engineering item and not an analytical blocker.

References

- [1] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s Property (T)*, New Math. Monographs **11**, Cambridge University Press, 2008.
- [2] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.
- [3] N. Bizi, C. Brouder, and F. Besnard, *Space and time dimensions of algebras with application to Lorentzian noncommutative geometry and quantum electrodynamics*, J. Math. Phys. **59** (2018), 062303; arXiv:1611.07062.
- [4] M. Bożejko and G. Fendler, *Herz–Schur multipliers and uniformly bounded representations of discrete groups*, Arch. Math. **57** (1991), 290–298.
- [5] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18; arXiv:gr-qc/9505009.
- [6] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.
- [7] A. Devastato, F. Lizzi, and P. Martinetti, *Time as an emergent property in spectral physics: gauge theories from spectral noncommutative geometry*, J. Math. Phys. **59** (2018), 112301.
- [8] C. Farsi and F. Latrémolière, *Continuity of spectral propinquity for spectrally truncated spheres*, preprint, 2024.

- [9] C. Farsi and F. Latrémolière, *Continuity for the spectral propinquity of the Dirac operators associated with an analytic path of Riemannian metrics*, arXiv:2504.11715, 2025.
- [10] N. Franco and M. Eckstein, *An algebraic formulation of causality for noncommutative geometry*, Class. Quantum Grav. **30** (2013), 135007; arXiv:1212.5171.
- [11] R. Geroch, *Topology in general relativity*, J. Math. Phys. **8** (1967), 782–786.
- [12] J. B. Hartle and S. W. Hawking, *Path-integral derivation of black-hole radiance*, Phys. Rev. D **13** (1976), 2188–2203.
- [13] E. Hekkelman and E. McDonald, *Spectral truncations of $\mathbb{T}^d$ and quantum metric geometry*, J. Noncommut. Geom., to appear (2024); arXiv:2403.18619.
- [14] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal., to appear (2025); arXiv:2412.00628.
- [15] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004.
- [16] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **415** (2023), Paper No. 108876, 88pp; preprint arXiv:1811.10843, November 2018.
- [17] F. Latrémolière, *The Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **370** (2018), 365–411.
- [18] L. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations of the torus*, Adv. Math. **439** (2024), Paper No. 109496.
- [19] A. Mondino and C. Sämann, *Lorentzian metric measure spaces*, Lett. Math. Phys. **114** (2024), Paper No. 37; arXiv:2209.14384.
- [20] A. Mondino and C. Sämann, *Synthetic Lorentzian Gromov–Hausdorff convergence and pre-compactness*, arXiv:2504.10380, 2025.
- [21] M. Che, R. Perales, and C. Sormani, *Gromov’s compactness theorem for the intrinsic timed-Hausdorff distance*, arXiv:2510.13069, October 2025.
- [22] G. Nieuviarts, *Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple*, arXiv:2502.18105 (v3, May 2025).
- [23] G. Nieuviarts, *Emergence of time from a twisted spectral triple in almost-commutative geometry*, arXiv:2512.15450, December 2025 (revised May 2026).
- [24] V. I. Paulsen, *Completely Bounded Maps and Operator Algebras*, Cambridge Studies in Advanced Mathematics 78, Cambridge University Press, 2002.
- [25] J.-P. Pier, *Amenable Locally Compact Groups*, Wiley, 1984.
- [26] G. Pisier, *Similarity Problems and Completely Bounded Maps*, Lecture Notes in Math. 1618, Springer, 2nd ed., 2001.
- [27] G. L. Sewell, *Quantum fields on manifolds: PCT and gravitationally induced thermal states*, Ann. Phys. (NY) **141** (1982), 201–224.
- [28] A. Strohmaier, *On noncommutative and pseudo-Riemannian geometry*, J. Geom. Phys. **56** (2006), 175–195.

- [29] M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979.
- [30] M. Toyota, *Quantum Gromov–Hausdorff convergence of spectral truncations for groups with polynomial growth*, arXiv:2309.13469, 2023.
- [31] W. G. Unruh, *Notes on black-hole evaporation*, Phys. Rev. D **14** (1976), 870–892.
- [32] K. van den Dungen, *Krein spectral triples and the fermionic action*, Math. Phys. Anal. Geom. **19** (2016), Art. 4, 22pp; arXiv:1505.01939.
- [33] J. de Groot, *Pseudo-Riemannian Spectral Triples for $SU(1,1)$* , arXiv:2601.22171, January 2026.
- [34] J. Loutey, *Bargmann–Segal lattice for the 3D harmonic oscillator and the Coulomb/HO structural asymmetry*, GeoVac internal preprint (2026), DOI via Zenodo.
- [35] J. Loutey, *The GeoVac spectral triple, the master Mellin engine case-exhaustion theorem, and the GH-convergence theorem on S^3* , GeoVac internal preprint, papers/synthesis (2026), DOI via Zenodo.
- [36] J. Loutey, *Latrémolière propinquity convergence of truncated Camporesi–Higuchi spectral triples on S^3* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [37] J. Loutey, *Tensor-product propinquity convergence for distinct-focal-length truncated Camporesi–Higuchi spectral triples on S^3* , GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [38] J. Loutey, *Latrémolière propinquity convergence of spectral truncations of compact Lie groups with bi-invariant metric*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [39] J. Loutey, *Tomita–Takesaki modular structure on truncated $SU(2)$ spectral triples; four-witness Wick-rotation literal identification at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [40] J. Loutey, *Lorentzian extension of the four-witness Wick-rotation theorem at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [41] J. Loutey, *Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.
- [42] J. Loutey, *Strong-form Lorentzian propinquity convergence on truncated $SU(2) \times U(1)_T$ Krein spectral triples*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Companion paper closing the strong-form gap identified as Q1/G1 of the present paper; identified the envelope refinement to (21) in Remark 4.5.)
- [43] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on $S^3 \times \mathbb{R}_t$ and the two-rate hybrid limit*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Closes the de-compactification limit G2 of the present paper at the norm-resolvent / spectral level via a two-rate hybrid composite; propinquity / metric level on a non-compact carrier remains open as named G2-metric / G1-genuine.)